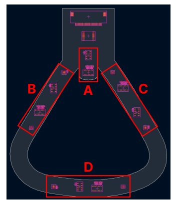
Notes on I2C Bus Groupings

As this board is both space-constrained and an unusual shape, components that are physically close on the board share I2C buses.
The general physical location of each group is illustrated to the right.



(Placement may differ)

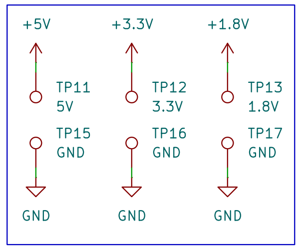
I2C Address Table

Bus Group	MS5611x	MAX30101x	TMP117x	SHT40x
A	0x77	0x57	–	–
B	0x77	0x57	0x48	0x44
C	0x77	0x57	0x48	0x44
D	0x77	0x57	0x48	0x44

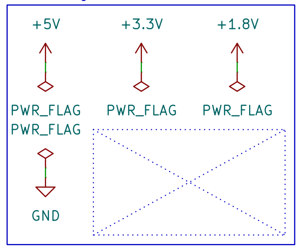
Switch Mapping

SDx/SCx	Bus
0	B
1	A
2	D
3	C

Test Points, Power

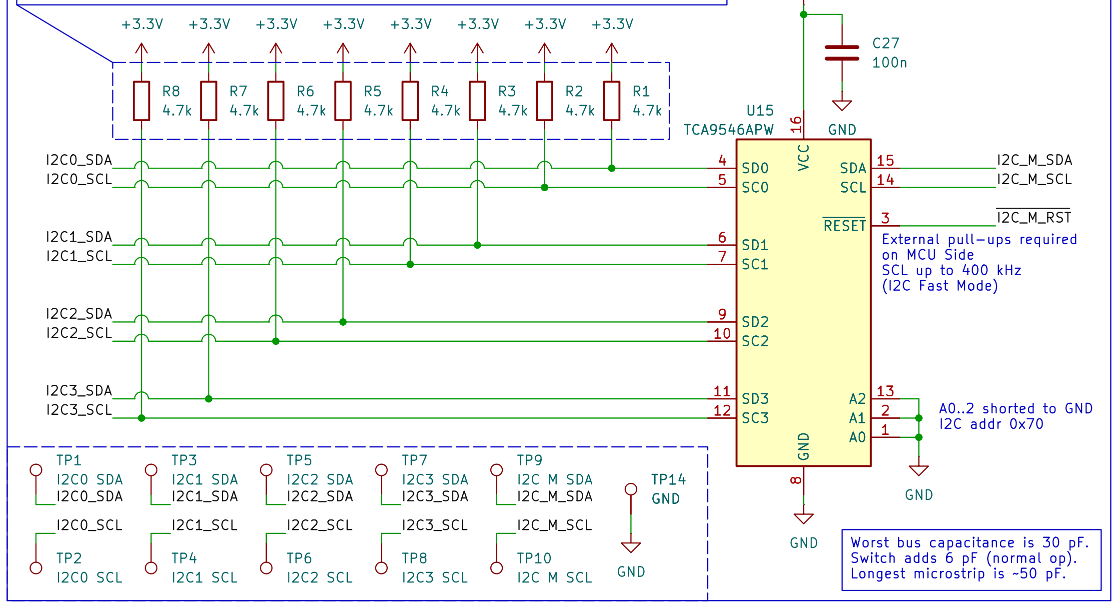


Power Flags



1:4 I2C Bus Switch

Note that this component is not a repeater.
Enabled channels add bus capacitance and pull-up resistors. Assuming the master-side I2C pull-ups are 4.7 kΩ, the total pull-up resistance when one channel is enabled is 2.35 kΩ.



Worst bus capacitance is 30 pF.
Switch adds 6 pF (normal op).
Longest microstrip is ~50 pF.

Notes on Power Supply

Maximum Current Consumption per Power Rail, Sensors

Part #	Quantity	1.8 V	3.3 V	5.0 V
MS5611x	4	–	1.4 mA	–
MAX30101x	4	1.1 mA	–	150 mA
TMP117x	3	–	0.3 mA	–
SHT40x	3	–	100 mA	–
Totals	–	4.4 mA	306.5 mA	600 mA

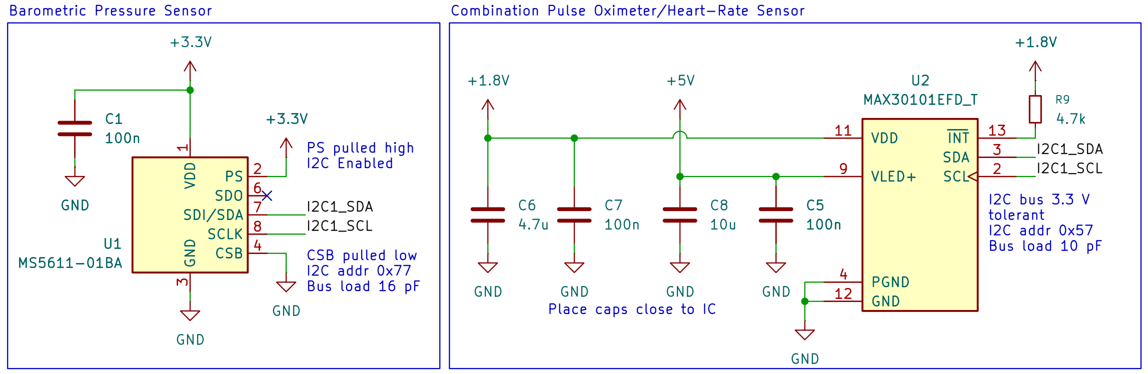
Maximum Current Consumption per Power Rail, Other

Part #	Quantity	1.8 V	3.3 V	5.0 V
TCA9546APW	1	–	<0.1 mA	–

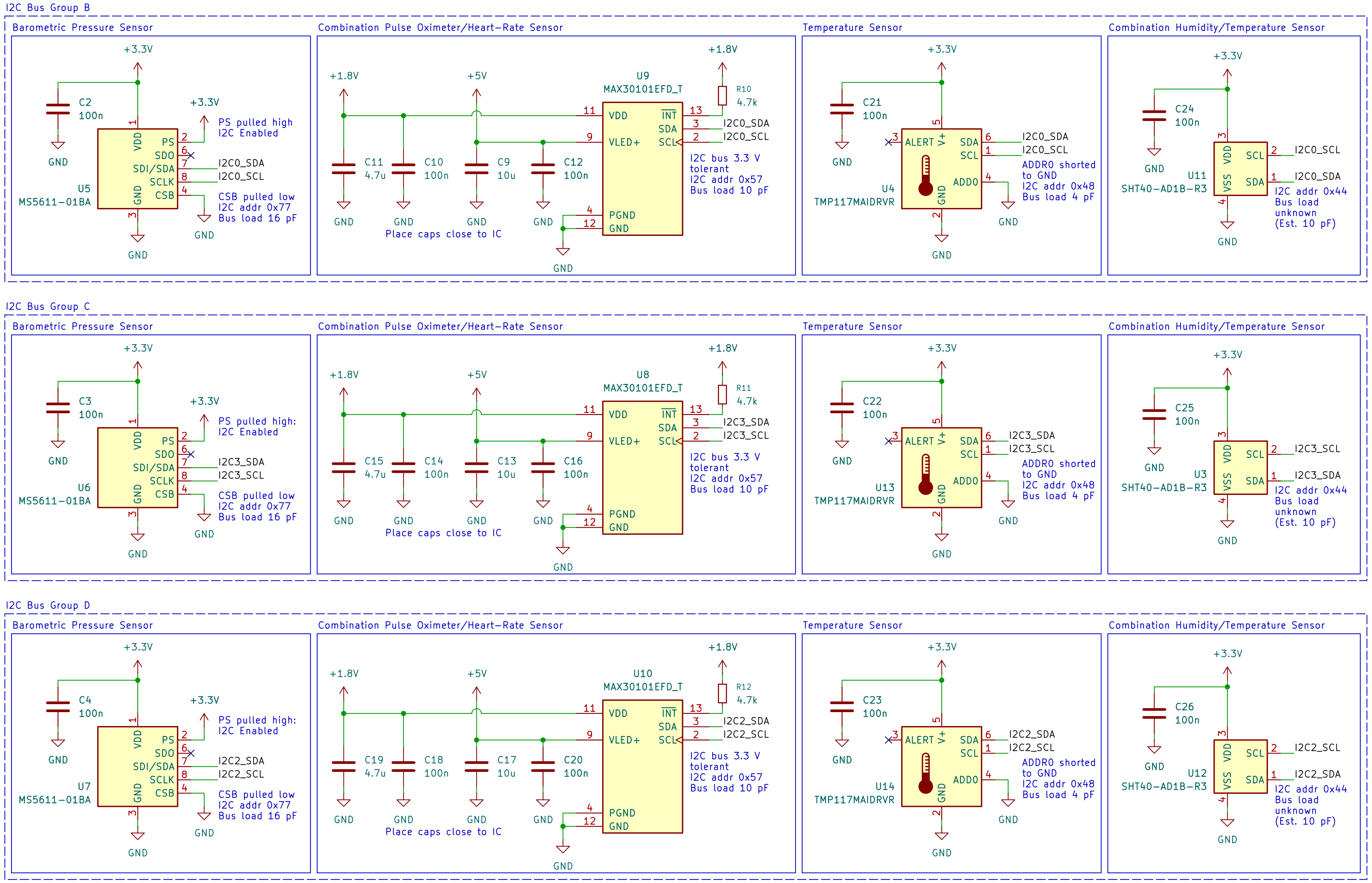
Recommended Supply Current per Power Rail, Entire Board

Condition	1.8 V	3.3 V	5.0 V
Recommended Minimum	5 mA	350 mA	800 mA
Maximum	10 mA	400 mA	1000 mA

I2C Bus Group A



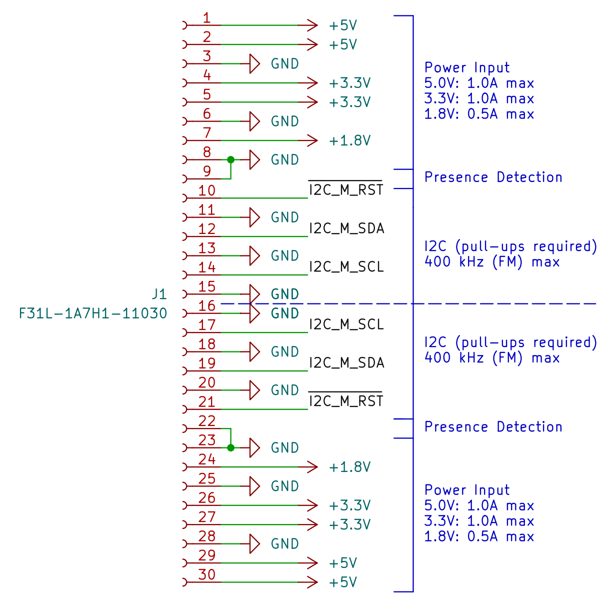
I2C Bus Group B



External FFC Connection

Total Power Input Capability:
5.0V: 2.0A max
3.3V: 2.0A max
1.8V: 1.0A max

Current Capacity Usage:
5.0V: 50%
3.3V: 20%
1.8V: <10%



Since it is very easy to accidentally use the wrong type of FFC cable, this connector is designed to be symmetrical so either type works.
Presence detection pins are also provided to detect when the cable is connected.

Mechanical

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

Aiden Lee [piman PCB]
Ka Moamoa Lab, Georgia Institute of Technology
Sheet: /
File: kmm-pmask-mask.kicad_sch

Title: Flexible Patient Monitoring Mask Insert

Size: A3	Date: 2026-07-15	Rev: 2.0
KiCad E.D.A. 10.0.4		Id: 1/1